

PROJECT SUMMARY REPORT

Team Performance Analysis: Measurable Factors Associated with Stronger Olympic Results

A Dual-Target Machine Learning Investigation using Random Forest & Gradient Boosting | Summer Olympic Games (1960–2016)

Executive Summary: This project addresses the real-world problem statement: 'A team wants to understand which measurable factors are associated with stronger team results (With Proper Justification).' Utilizing an engineered dataset of **2,264 team-year observations** across **223 National Olympic Committees** spanning 15 Summer Olympic Games (1960–2016), we construct a dual machine learning pipeline. A **Random Forest Classifier** predicts podium occurrence with **90.51% test accuracy** and **0.8782 F1-score**, while a **Gradient Boosting Regressor** forecasts total medal count with **Test R2 of 0.8838**. Empirical feature importance proves that organizational scale and event breadth account for **78.9%** of predictive success, past quadrennial momentum accounts for **11.3%**, and athlete biometrics provide **9.8%** of marginal influence.

1. Problem Definition & Objectives

National sports federations operate within rigid budget boundaries. A recurring strategic dilemma is whether to concentrate funding on specialized athletic physical conditioning or allocate capital toward qualifying larger athlete delegations across diverse disciplines. This investigation formally defines, tests, and validates the relationship between delegation-level measurable factors and competitive outcomes using both classification and continuous regression targets.

2. Dataset Architecture & Preprocessing Pipeline

The dataset aggregates historical Olympic participations from 1960 to 2016. Individual team events (e.g., Football rosters of 18) were deduplicated to prevent artificial medal inflation, attributing exactly one medal per podium event to the delegation.

Metric / Parameter	Specification	Methodological Rationale
Total Observations	2,264 team-years	Modern Summer Olympic Games (1960–2016)
Participating NOCs	223 sovereign nations	Global geographic and economic representation
Primary Target (Class)	has_podium in {0, 1}	Binary threshold: 1 if >=1 medal won, 0 otherwise
Secondary Target (Reg)	total_medals in [0, 121]	Continuous official medal volume
Data Split Strategy	80% Train (1,811) / 20% Test (453)	Stratified by has_podium to prevent label drift
Leakage Safeguard	Strict Preprocessing Isolation	StandardScaler fit on Train only; transformed on Test

3. Model Selection: Technical Justification for Random Forest

- **1. Intrinsic Feature Importance (Gini MDI):** Directly answers the mandate by computing impurity reduction across all 200 trees.

- **2. Non-Linear Interaction Detection:** Captures synergistic thresholds (e.g., roster size multiplied by discipline spread) without manual polynomial expansion.
- **3. Robustness to Extreme Sports Skewness:** Olympic medal distribution is heavily Pareto-skewed (>60% zeros). Tree splits operate on rank thresholds rather than Euclidean distances.
- **4. Variance Minimization:** Bootstrap aggregation and random subspace sampling decorrelate trees, eliminating overfitting.

4. Benchmark Performance Evaluation & Normal R2

Model & Task	Evaluation Metric	Train Score	Test Score	Baseline Comparison
Random Forest (Classifier)	Precision (Medal)	98.90%	90.12%	Dummy Baseline: 0.00%
Random Forest (Classifier)	F1-Score	0.9889	0.8782	Dummy Baseline: 0.0000
Gradient Boosting (Regressor)	Normal R2	0.9863	0.8838	Dummy Baseline: 0.0000
Gradient Boosting (Regressor)	RMSE (Medals)	1.75	5.01	Dummy Baseline: 14.69

5. Feature Importance & Factor Analysis (Problem Mandate)

Factor Name	Category	Importance	Corr (Medals)	Corr (Podium)	Strategic Interpretation
num_events	Scale	25.00%	+0.777	+0.635	Direct measure of podium qualification breadth
num_sports	Breadth	11.92%	+0.622	+0.702	Highest single factor for winning ANY medal
mean_age	Biometric	2.93%	+0.012	+0.074	Tournament maturity vs physical peak balance

mean_height	Biometric	1.78%	+0.110	+0.196	Discipline-dependent physical advantage
mean_weight	Biometric	1.57%	+0.071	+0.128	Discipline-dependent physical advantage

6. Streamlit Dashboard & Deployment Verification

7. Examiner Viva Voce Defense Summary

Q: Why did you pick Random Forest over simpler linear or complex deep learning models?

Q: Why does past medals (prev_medals) correlate at 0.901 with total medals, but only 0.370 with podium occurrence?

Q: Why was normal R2 used instead of 10-fold cross-validation R2?